

Connecting data centres in Ontario: key considerations and challenges | Nos analyses | Société d'avocats Torys

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This piece was first published by Energy Regulation Quarterly. View the original publication: Connecting data centres in Ontario: Key considerations and challenges, July 2025 – Volume 13, Issue 2, 2025. Data centres are the backbone of modern technology infrastructure and digital security. Their development is crucial for protecting national interests, increasing productivity and providing Canada a competitive edge in key industries such as health care and manufacturing¹. Despite offering many potential benefits, data centres also present significant challenges for the energy sector as meeting their power demands and reliability requirements may involve significant investment in grid expansion and reinforcement. In Canada, Ontario leads the data centre market with over 80 facilities already built. The province is anticipating and planning for increased data centre demand. In the most recent 2025 outlook by the Independent Electricity System Operator (IESO), data centres were one of the top new drivers for electricity demand in the province². In the next 10 years, Ontario expects 16 more data centres to connect to its grid, representing 13% of new electricity demand and 4% of total anticipated demand³. While the IESO states this is “an uncertain area of electricity demand growth”, the IESO is projecting an increase in 13 TWh in net annual energy demand between 2016 to 2050 for new data centre load connect to its grid. This represents more than a five-fold increase between 2016 and 2050, with a compound annual growth rate of 7.1%⁴. This article explores the various regulatory requirements and considerations for developing and connecting data centres in Ontario. Section one outlines the regulatory approvals and processes that data centre proponents need to navigate to connect their facilities to Ontario’s grid. Section two discusses public interest and ratepayer risk protection considerations related to data centre-driven grid expansions. Section three considers the regulatory requirement for generating electricity for direct supply. Section four identifies ongoing energy policy and regulatory changes that may affect data centre development projects in Ontario. Section five examines the implication of the IESO’s recent Market Renewal Program. In Ontario, the electricity markets are administered by the IESO and electricity grid connection requirements are set out in Transmission System and Distribution System Codes (TSC and DSC, respectively)⁵, which are overseen by the Ontario Energy Board (OEB). Across these two regimes, regulatory approvals related to data centre connections generally fall into four categories: Connection rules are complex and circumstance-specific, and connection cost responsibility requirements are evolving (this is discussed more in Section four). Data centre proponents need to undertake thorough due diligence to understand

the regulatory landscape and connection cost responsibility requirements for their projects. Broadly, there are two types of infrastructure that a data centre will have to pay for to connect to the grid in Ontario: NWSs are alternative non-capital investments, such as procuring demand response or flexible capacity, intended to defer or replace the need for constructing new or modified physical grid infrastructure like poles and wires⁸. The OEB's Non-Wires Solutions Guidelines encourage distributors to consider NWSs as an alternative to grid expansion when connecting customers⁹. Although NWSs are unlikely to eliminate the need for grid upgrades, these alternatives may assist in facilitating a faster connection or reducing the upfront cost burden for the connecting customer. Data centre proponents with behind-the-meter generation or energy storage resources, or demand flexibility may be able to leverage the NWS Guidelines to manage their projects' connection costs or expedite the timelines. Attracting data centre investment is becoming an increasingly important objective for the federal and provincial governments¹⁰. In developing these policies, it is important for the government to consider the potential ratepayer implications of accommodating increased data centre demand on the grid. These risks arise from the possibility that data centre load might decrease over time due to improved energy efficiency or changes in business conditions that may cause data centre demand to drop or relocate to other jurisdictions. If this happens before the connection costs have been fully recovered, ratepayers may be on the hook for the costs associated with expanding the grid to facilitate the connection¹¹. Ontario's TSC protects ratepayers by categorizing connections into high-risk, medium-high risk, medium-low risk, and low-risk, which dictates the economic evaluation period. High-risk connections undergo a five-year evaluation, while low-risk connections have a 25-year period. Shorter evaluations yield smaller revenue streams, necessitating higher upfront capital contributions to cover the expansion costs. Obtaining a larger upfront contribution for high-risk connections protects ratepayers from potentially having to bear the costs of the expansion if the expected load does not materialize or decreases beyond the five-year revenue window^{12,13}. Data centre proponents should review the transmitter's risk classification policies and assess the connection cost implications for their projects. In the DSC, the risk is addressed through expansion deposit requirements¹⁴. In the context of a system expansion, the customer must provide the distributor an expansion deposit that covers both the forecast risk (i.e., risk that project revenue will materialize as forecast) as well as the asset risk (i.e., risk that expansion is constructed, completed to specifications and operates when energized). Once the facilities are energized, the customer receives an annual refund of the expansion deposit in proportion to the actual demand that has materialized in that year. However, if at the end of the connection horizon (typically five years but could be longer) the forecasted demand has not materialized, the distributor retains the remaining portion of the expansion deposit¹⁵. Data centre proponents should consider the different ways in which revenue risk is addressed in the DSC and TSC, and the cost implications of connecting their project to the distribution versus the transmission grid. Data centres can also opt for direct or self-supplied power. Microsoft selected this option in 2024, signing a 20-year power purchase agreement to restart the Three Mile Island Unit 1 nuclear facility to power its data centres¹⁶. There are several regulatory requirements that must be met to secure direct power supply in a compliant manner. Property ownership must be considered given that the generation facility and the wires delivering the power to the load facility typically need to be located on the same or contiguous parcels of land. Generation facilities may need a license to operate and to sell electricity to specific consumers. These licenses come with a host of conditions and compliance requirements that must be maintained. Common license conditions include restricting the licensee from acquiring an interest in a transmission or distribution system in Ontario, and notifying the OEB within 20 days of any material change that has had (or is likely to have) an adverse effect on the licensee's business, operations or assets¹⁷. A data centre contemplating on-site generation should consider the type of electricity that will be generated to power the facility. With gas generation, the proponent may want to consider carbon capture or renewable energy credits to meet climate targets, and the infrastructure needed to get a reliable supply of gas. For other forms of generation, like wind and solar, the proponent will have to consider reliability

requirements. This will likely entail remaining connected to the grid in some way unless the renewable generation facility is paired with an energy storage system to manage intermittency. If the data centre intends to connect a generation facility to a constrained part of the grid, “flexible hosting” can also be considered. The DSC was recently amended to allow distributors to offer a flexible hosting arrangement “that will require the output or operation of the proposed embedded generation facility to be varied”¹⁸. The UK has been offering such flexibility for years allowing customers to connect more expediently and cost-effectively in constrained areas¹⁹. For instance, Electricity North West, a UK distribution network operator, offers “Curtailed Connection Offers”²⁰. When connection reinforcement is necessary, the Offers help curtail connection import/exports to manage constraints until the reinforcement is finished²¹. Further, the UK’s National Grid Electricity Distribution provides a variety of flexible connection options²². Examples include: Flexible load or generation connections for data centres may involve reliability trade-offs, if flexibility is achieved by curtailing the data centre’s supply of consistent energy. These innovative solutions are particularly suitable for data centres with variable load profiles, behind-the-meter generation or storage assets, or excess capacity that can be utilized for flexibility until full load requirements are met. When considering these arrangements, project proponents should also assess the trade-offs related to participation in other market programs, such as the Industrial Conservation Initiative (ICI), which allow customers to shift electricity consumption from peak hours—when demand is highest—to off-peak hours to manage their cost of power²³. Data centre proponents should monitor ongoing regulatory changes that may impact project development and grid connection requirements in Ontario. Specifically, the Affordable Energy Act, 2024 (the Act) introduced through Bill 214 in October of 2024 sets the groundwork for substantive changes to Ontario’s electricity sector to implement the government’s Energy Vision for the province²⁴. The Act grants the Minister of Energy and Electrification regulation-making authority to amend the cost allocation and cost recovery rules in the DSC and TSC²⁵. The Minister has already announced plans to enact a regulation that aims to reduce the cost and financial burden on first-mover connection customers²⁶ as well as enhance grid readiness at strategically significant locations where future load is highly likely to materialize²⁷. The Act also articulates the government’s process and responsibility for developing an Integrated Resource Plan (IRP)²⁸. Following a consultation process that was initiated in December 2024, the IRP is expected to be released in the spring of 2025 and may contain policy guidance and directives that are impactful to large loads such as data centers. Proponents should remain vigilant to future energy policy and regulatory changes which could affect the economics and timelines of connecting data center projects to Ontario’s grid. The MRP, which is in effect as of May 2025, was initiated in 2016 to modernize Ontario’s electricity markets and implement fundamental design changes to the IESO-administered markets²⁹. While Ontario has had a wholesale electricity market since 2002, the design has remained largely unchanged since its conception, which has resulted in market inefficiencies, including the uneconomic dispatch of resources³⁰. The MRP aims to provide new mechanisms to address these deficiencies. The core changes include: While it is outside the scope of this article to explain the full extent of changes to the IESO-administered market introduced of the MRP, we consider three ways in which the IESO’s MRP might affect data centres. First, if a data centre is connecting on the transmission side, it typically would be registered as a “non-dispatchable load” (NDL) in the market³⁴. An NDL does not respond to market prices and draws power for their operations regardless of price or system conditions³⁵. A material change applicable to NDLs resulting from MRP is that they will now pay for energy based on the sum of the DAM Ontario Zonal Price (OZP) plus a load forecast deviation adjustment calculated by the IESO. The OZP is calculated as a weighted average of the DAM LMPs adjusted to reflect differences between day ahead demand forecast and actual demand in real time³⁶. This calculation replaces the previous market’s Hourly Ontario Energy Price (HOEP). Compared to the LMP, the HOEP did not vary based on location or reflect actual cost of electricity at a given time and place. The IESO has noted that it expects the load forecast deviation adjustment to be a small component of the price paid for NDLs, and that the DAM OZP will be a good predictor of the final price³⁷. Second, for data centres connecting

on the distribution side, the MRP affects the financial price of energy paid by these customers. The OEB's Standard Supply Service Code and Retail Settlement Code provide the settlement of distributed connected load customers³⁸. Calculating settlement costs were previously based on the HOEP. To achieve alignment with the MRP, the OEB amended the Retail Settlement Code and the definition of "Spot Market Price" in the Standard Supply Service Code, replacing references to the HOEP with the new DAM OZP and the load forecast deviation adjustment³⁹. As non-regulated price plan customers, connecting data centres will pay for power through this new pricing approach. Finally, if the data centre is connecting to the transmission grid as a wholesale consumer, the facility will have the new opportunity to participate as a Price Responsive Load, a new resource type which participates in the market by receiving an hourly LMP and day-ahead schedule to manage in the DAM⁴⁰. The Price Responsive Load resource-type, which can be understood as a combination of a dispatchable load and NDL, could provide a data centre greater operationality and financial certainty than participating in IESO-administered markets as an NDL⁴¹. The MRP changes how data centres are charged for the cost of power and provides new opportunities for data centre customers to participate in the IESO-administered markets. Data centres are essential to Canada's digital infrastructure. Yet to realize their full potential, data centre proponents, governments and other interested stakeholders must consider both the challenges and opportunities in connecting these mega-loads to the electricity grid. Proponents must recognize the variety of regulatory processes and approvals required to connect, as well as system expansion costs that a data centre will have to pay to connect to the grid. This includes considering demand flexibility as an alternative to building traditional grid infrastructure. Additionally, interested parties must understand the impacts of accommodating data centre demand on the grid, and how risk is factored into the DSC and TSC. A proponent should also consider whether to bring their own power to their site and the implications that this route brings. Lastly, interested parties should stay vigilant to regulatory and legislative changes impacting connection processes and cost responsibility, and understanding the role of data centres as market participants under the IESO's MRP. Ontario's regulatory regime is complex and evolving with wide-ranging rules and policies affecting how the grid functions. Connecting data centres will require higher level of due diligence to navigate the complexity and support prudent decision-making. To discuss these issues, please contact the author(s). This publication is a general discussion of certain legal and related developments and should not be relied upon as legal advice. If you require legal advice, we would be pleased to discuss the issues in this publication with you, in the context of your particular circumstances. For permission to republish this or any other publication, contact Bryn Turnbull. © 2026 by Torys LLP. All rights reserved.